Peter Carter

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Education

North Carolina State University Raleigh, NC

B.S. Computer Science, AI concentration, Minor in Mathematics *Expected December 2026*

- Relevant Coursework: Data Structures and Algorithms, Operating Systems, Software Engineering, Database Management Systems, Artificial Intelligence, Automated Learning and Data Analysis, Computer Organization and Assembly, Senior Design

Technical Skills

Languages: Java, Python, C, JavaScript, TypeScript, SQL
Frameworks and Libraries: React, Next.js, Spring Boot, Hibernate, NestJS, Prisma, PyTorch, scikit-learn, CatBoost, pandas, NumPy, Vite
Databases: PostgreSQL, MySQL
Developer Tools: Docker, VS Code, Eclipse IDE, Cursor IDE, Jenkins, GitHub, JUnit, Maven, Vercel, Figma

Projects

Clarify | *React, Next.js, TypeScript, NestJS, Prisma, PostgreSQL, Docker* clarify-recipes.com

- Built and deployed a full-stack recipe manager that imports recipes from a pasted URL by parsing schema.org JSON-LD data, skipping ads and unrelated content; supports tagging, search, manual editing, and calorie/macro estimates
- Reduced recipe-import latency by ˜90% (from ˜5s to under 0.5s for previously-seen ingredients) by re-architecting third-party API calls to a constant 2 requests per ingredient (down from up to 6) and adding a response cache
- Covered the backend with 158 automated tests (131 unit, 27 end-to-end), reaching 80%+ statement/branch coverage on core business logic

Accessible Periodic Table | *React, Next.js, JavaScript, HTML, Amazon Polly | In Progress*

- Building an interactive periodic table for blind and low-vision students as a senior capstone sponsored by College Board, in a team of five
- Implemented four-direction arrow-key navigation that preserves the table's spatial layout, plus element search, bookmarks for side-by-side comparison, and a high-contrast mode
- Replaced screen-reader pronunciation errors (e.g., "Mg" read as "milligrams") with custom Amazon Polly narration; built an AP-compliant testing mode and gathering user feedback

Jaw-Clench Controlled Robot | *Python, scikit-learn | BR41N.IO Spring School 2026, 2nd Place*

- Trained a random forest classifier on ˜200 self-collected EEG samples from a NeuroPawn Knightboard to detect jaw-clench patterns in real time
- Mapped clenches to forward and rotation commands sent wirelessly to a Sphero BOLT robot, placing 2nd among 42 attendees from 3 universities

WolfCafe | *Java, Spring Boot, Hibernate, MySQL, React, JavaScript*

- Built the backend for a full-stack cafe point-of-sale system, designing REST APIs in Java/Spring Boot for order management, inventory tracking, and staff roles, with 80%+ JUnit coverage across backend logic
- Modeled a MySQL database with Hibernate and integrated it with a React/JavaScript frontend

Experience

Software Team Lead August 2025 - Present

NCSU Neurotech Club *Raleigh, NC*

- Built a custom visualizer for NeuroPawn Knightboard EEG hardware with 8-channel displays, a built-in hardware-free simulator, an SSVEP calibration/testing/recognition suite, and recording export, shipped for web and desktop
- Secured a research partnership with Synaptive, an EEG dry-electrode company, to collect data and test electrodes designed for different hair types
- Designed and built the club's website